Session 15 (AIML) - Assignment Questions

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Branch: **Artificial Intelligence and Machine Learning (AIML)**
Domain: **AIML (Artificial Intelligence and Machine Learning)**

Github Repo Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments

Github File Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments/tree/main/Assignment15

Dataset Used: Kaggle – Countries of the World Dataset
https://www.kaggle.com/datasets/fernandol/countries-of-the-world

Program Output:

1. Import the Pandas library.

```
[1]: """ ASSIGNMENT 15 """

     '''
     Countries Dataset - Data Reading, Loading, Exploration, and Understanding
     '''
```

```
[1]: '\nCountries Dataset - Data Reading, Loading, Exploration, and Understanding\n'
```

```
[2]: # Importing Libraries
     import pandas as pd
     import numpy as np
```

2. Load the CSV dataset into a DataFrame.

```
[18]: # Load the dataset

      df = pd.read_csv("countries of the world.csv")
```

3. Display the first 5 rows of the dataset.

```
[20]: # Display first 5 rows

      df.head()
```

[20]:

	Country	Region	Population	Area (sq. mi.)	Pop. Density (per sq. mi.)	Coastline (coast/area ratio)	Net migration	Infant mortality (per 1000 births)	GDP ($ per capita)	Literacy (%)	Phones (per 1000)	Arable (%)	Crops (%)	Other (%)	Climate	Birthrate	Deathrate	Agricultur
0	Afghanistan	ASIA (EX. NEAR EAST)	31056997	647500	48,0	0,00	23,06	163,07	700.0	36,0	3,2	12,13	0,22	87,65	1	46,6	20,34	0,3
1	Albania	EASTERN EUROPE	3581655	28748	124,6	1,26	-4,93	21,52	4500.0	86,5	71,2	21,09	4,42	74,49	3	15,11	5,22	0,23
2	Algeria	NORTHERN AFRICA	32930091	2381740	13,8	0,04	-0,39	31	6000.0	70,0	78,1	3,22	0,25	96,53	1	17,14	4,61	0,10
3	American Samoa	OCEANIA	57794	199	290,4	58,29	-20,71	9,27	8000.0	97,0	259,5	10	15	75	2	22,46	3,27	Na
4	Andorra	WESTERN EUROPE	71201	468	152,1	0,00	6,6	4,05	19000.0	100,0	497,2	2,22	0	97,78	3	8,71	6,25	Na

4. Display the last 5 rows of the dataset.

```
[22]: # Display last 5 rows

df.tail()
```

[22]:

	Country	Region	Population	Area (sq. mi.)	Pop. Density (per sq. mi.)	Coastline (coast/area ratio)	Net migration	Infant mortality (per 1000 births)	GDP ($ per capita)	Literacy (%)	Phones (per 1000)	Arable (%)	Crops (%)	Other (%)	Climate	Birthrate	Deathrate	Agriculture
222	West Bank	NEAR EAST	2460492	5860	419,9	0,00	2,98	19,62	800.0	NaN	145,2	16,9	18,97	64,13	3	31,67	3,92	0,09
223	Western Sahara	NORTHERN AFRICA	273008	266000	1,0	0,42	NaN	NaN	NaN	NaN	NaN	0,02	0	99,98	1	NaN	NaN	NaN
224	Yemen	NEAR EAST	21456188	527970	40,6	0,36	0	61,5	800.0	50,2	37,2	2,78	0,24	96,98	1	42,89	8,3	0,135
225	Zambia	SUB-SAHARAN AFRICA	11502010	752614	15,3	0,00	0	88,29	800.0	80,6	8,2	7,08	0,03	92,9	2	41	19,93	0,22
226	Zimbabwe	SUB-SAHARAN AFRICA	12236805	390580	31,3	0,00	0	67,69	1900.0	90,7	26,8	8,32	0,34	91,34	2	28,01	21,84	0,179

5. Show the number of rows and columns.

```
[23]: # Display number of rows and columns

print("Shape:", df.shape)
```

```
Shape: (227, 20)
```

6. Display all column names.

```
[9]: # Display column names
print(df.columns)
```

```
Index(['Country', 'Region', 'Population', 'Area (sq. mi.)',
       'Pop. Density (per sq. mi.)', 'Coastline (coast/area ratio)',
       'Net migration', 'Infant mortality (per 1000 births)',
       'GDP ($ per capita)', 'Literacy (%)', 'Phones (per 1000)', 'Arable (%)',
       'Crops (%)', 'Other (%)', 'Climate', 'Birthrate', 'Deathrate',
       'Agriculture', 'Industry', 'Service'],
      dtype='object')
```

7. Show dataset information, data types, and non-null values.

```
[10]: # Display dataset information
      df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 227 entries, 0 to 226
Data columns (total 20 columns):
 #   Column                              Non-Null Count  Dtype  
---  ------                              --------------  -----  
 0   Country                             227 non-null    object 
 1   Region                              227 non-null    object 
 2   Population                          227 non-null    int64  
 3   Area (sq. mi.)                      227 non-null    int64  
 4   Pop. Density (per sq. mi.)          227 non-null    object 
 5   Coastline (coast/area ratio)        227 non-null    object 
 6   Net migration                       224 non-null    object 
 7   Infant mortality (per 1000 births)  224 non-null    object 
 8   GDP ($ per capita)                  226 non-null    float64
 9   Literacy (%)                        209 non-null    object 
 10  Phones (per 1000)                   223 non-null    object 
 11  Arable (%)                          225 non-null    object 
 12  Crops (%)                           225 non-null    object 
 13  Other (%)                           225 non-null    object 
 14  Climate                             205 non-null    object 
 15  Birthrate                           224 non-null    object 
 16  Deathrate                           223 non-null    object 
 17  Agriculture                         212 non-null    object 
 18  Industry                            211 non-null    object 
 19  Service                             212 non-null    object 
dtypes: float64(1), int64(2), object(17)
memory usage: 35.6+ KB
```

8. Generate summary statistics for numerical columns.

```
[11]: # Display summary statistics
      df.describe()
```

[11]:

	Population	Area (sq. mi.)	GDP ($ per capita)
count	2.270000e+02	2.270000e+02	226.000000
mean	2.874028e+07	5.982270e+05	9689.823009
std	1.178913e+08	1.790282e+06	10049.138513
min	7.026000e+03	2.000000e+00	500.000000
25%	4.376240e+05	4.647500e+03	1900.000000
50%	4.786994e+06	8.660000e+04	5550.000000
75%	1.749777e+07	4.418110e+05	15700.000000
max	1.313974e+09	1.707520e+07	55100.000000

9. Display the data type of each column.

```
[12]: # Display data types
      print(df.dtypes)
```

```
Country                                 object
Region                                  object
Population                               int64
Area (sq. mi.)                           int64
Pop. Density (per sq. mi.)              object
Coastline (coast/area ratio)            object
Net migration                           object
Infant mortality (per 1000 births)      object
GDP ($ per capita)                     float64
Literacy (%)                            object
Phones (per 1000)                       object
Arable (%)                              object
Crops (%)                               object
Other (%)                               object
Climate                                 object
Birthrate                               object
Deathrate                               object
Agriculture                             object
Industry                                object
Service                                 object
dtype: object
```

10. Count missing values in each column.

```
[13]: # Count missing values
      print(df.isnull().sum())
```

```
Country                                0
Region                                 0
Population                             0
Area (sq. mi.)                         0
Pop. Density (per sq. mi.)             0
Coastline (coast/area ratio)           0
Net migration                          3
Infant mortality (per 1000 births)     3
GDP ($ per capita)                     1
Literacy (%)                          18
Phones (per 1000)                      4
Arable (%)                             2
Crops (%)                              2
Other (%)                              2
Climate                               22
Birthrate                              3
Deathrate                              4
Agriculture                           15
Industry                              16
Service                               15
dtype: int64
```

11. Count unique values in each column.

```
[14]: # Count unique values
      print(df.nunique())
```

```
Country                               227
Region                                 11
Population                            227
Area (sq. mi.)                        226
Pop. Density (per sq. mi.)            219
Coastline (coast/area ratio)          151
Net migration                         157
Infant mortality (per 1000 births)    220
GDP ($ per capita)                    130
Literacy (%)                          140
Phones (per 1000)                     214
Arable (%)                            203
Crops (%)                             162
Other (%)                             209
Climate                                 6
Birthrate                             220
Deathrate                             201
Agriculture                           150
Industry                              155
Service                               167
dtype: int64
```

12. Display 5 random rows from the dataset.

```
[15]: # Display 5 random rows
      df.sample(5)
```

[15]:

	Country	Region	Population	Area (sq. mi.)	Pop. Density (per sq. mi.)	Coastline (coast/area ratio)	Net migration	Infant mortality (per 1000 births)	GDP ($ per capita)	Literacy (%)	Phones (per 1000)	Arable (%)	Crops (%)	Other (%)	Climate	Birthrate	Deathrate	Agriculture
19	Belgium	WESTERN EUROPE	10379067	30528	340,0	0,22	1,23	4,68	29100.0	98,0	462,6	23,28	0,4	76,32	3	10,38	10,27	0,01
64	Estonia	BALTICS	1324333	45226	29,3	8,39	-3,16	7,87	12300.0	99,8	333,8	16,04	0,45	83,51	3	10,04	13,25	0,04
166	Qatar	NEAR EAST	885359	11437	77,4	4,92	16,29	18,61	21500.0	82,5	232,0	1,64	0,27	98,09	1	15,56	4,72	0,002
189	South Africa	SUB-SAHARAN AFRICA	44187637	1219912	36,2	0,23	-0,29	61,81	10700.0	86,4	107,0	12,08	0,79	87,13	1	18,2	22	0,025
74	Gaza Strip	NEAR EAST	1428757	360	3968,8	11,11	1,6	22,93	600.0	NaN	244,3	28,95	21,05	50	3	39,45	3,8	0,03

13. Count occurrences of unique values (for a selected column).

```
[16]: # Count occurrences of values in the Country column
      print(df["Country"].value_counts())
```

```
Country
Afghanistan       1
Albania           1
Algeria           1
American Samoa    1
Andorra           1
                 ..
West Bank         1
Western Sahara    1
Yemen             1
Zambia            1
Zimbabwe          1
Name: count, Length: 227, dtype: int64
```